5.0 COMPACT SUBSTATION REMOTE TERMINAL UNIT (CSRTU)

5.1 General requirements

5.1.1 This functional specification defines the requirements of Remote Terminal Units for SCADA-equipped Compact Substation only.

5.1.2 Compact Substations equipped with RTUs will provide the facility for Regional Control Centres (RCCs) to monitor and control Compact Substations in Distribution network.

5.1.3 A SCADA-equipped Compact Substation's RTU (CSRTU) acquires status inputs from protection relays, switchgears and ancillary equipment and sends real-time status updates to RCC besides providing the facilities for control commands from RCC to plant equipment for remote operations.

5.1.4 The CSRTU shall be designed for compact installation and located in a Secondary Enclosure Panel as specified in Secondary Enclosure Panel Section.

5.1.5 The connections for SCADA monitoring and control facilities for both SCADA-equipped Compact Substation to Remote Control Box and Remote Control Box to RTU shall be pre-wired at the factory.

5.2 Environmental Requirements

5.2.1 The CSRTU shall be classed as outdoor device and shall be suitable for continuous operation in Malaysia's tropical monsoon climate, which includes exposure to severe frequently occurring thunderstorms. It shall also be suitable for conditions in which they will be exposed to heavy industrial pollution, salt-spray, and high levels of airborne dust. Except for stainless steel, all steel surfaces that are not galvanized shall be treated to protect against corrosion.

5.2.2 Operating Temperature

The CSRTU shall be tested up to + 60°C following IEC 60068-2-2.

5.2.3 Relative Humidity

The CSRTU shall be tested up to 95% at 40°C following IEC 60068-2-30.

5.2.4 Cyclic Damp Heat and Absolute Humidity

The CSRTU shall be tested between +25°C to +40°C following IEC 60068-2-30.

5.2.5 Vibration and Shock

The CSRTU shall be tested following IEC 60068-2-6 Sinusoidal Vibration and IEC 60068-2-27 Shock/Bump methods.

5.3 Electrical Requirements

5.3.1 The CSRTU shall be designed for safe operation in high voltage environment. Data communication ports shall be tested to withstand disturbance test without permanent corruption of data, and subsequent delay of data transfer.

5.3.2 The following type test requirements shall apply to all modules covered under the project's scope of supply unless specifically stated otherwise.

5.3.3 Tests shall be performed on the CSRTU system (which means a complete functional CSRTU with all the processor, serial communications and input/output modules assembled) if the test specifically mentions it.

Requirement	Following Test Standard
Dielectric Test Voltage	IEC 60255-5
Impulse Voltage Test	IEC 60255-5
DC supply interruptions	IEC 60255-11
High Frequency Disturbance (Damped Oscillatory Waves)	IEC 60255-22-1 or IEC 61000-4-12
Electrostatic Discharge Immunity	IEC 61000-4-2
Radiated Electromagnetic Field Disturbance	IEC 61000-4-3
Fast Transient	IEC 61000-4-4
Surge Immunity	IEC 61000-4-5
Conducted disturbances induced by radio frequency fields, amplitude modulated	IEC 61000-4-6
Power Frequency Magnetic Field Immunity	IEC 61000-4-8

* The successful tenderer shall show proof that the equipment and all its modules are fully compliant to all type test requirements as defined in this specification.

The test reports can either be obtained from tests witnessed by TNB personnel or from laboratories which are accredited with IEC/ISO 17025.

The test reports shall be submitted to TNB, not later than 2 months after award of contract, failing which, TNB has the right to terminate the contract.

5.4 Design requirements

5.4.1 The CSRTU shall be equipped with a power supply module that shall provide all necessary power needed for:

(i) operation of the CSRTU itself
(ii) isolated power for the "wetting" of contact inputs
(iii) operation of the coils of heavy duty interposing relays of the control output points
(iv) a 12V auxiliary power ready for TNB-issued Modem/Data Transmission Unit (DTU) and its power requirement. The maximum power consumption of the modem is 5 Watt.

5.4.2 The CSRTU power supply module shall receive its input power from Compact Substation's DC supply system with nominal voltage of 24Vdc, with a tolerance of +20% and -15% of the nominal voltage and maintain full normal operation over the input supply range.

5.4.3 The CSRTU shall be equipped with all hardware and software necessary to record:

(i) CSRTU system events that describes the operation and status of the CSRTU (for example: initialization, start-ups, logins/logouts, configuration uploads/downloads, module failures and communication faults).
(ii) Changes in states of digital inputs

5.4.4 The CSRTU shall be designed to prevent false control actions from being executed and erroneous data from being transmitted due to communication channel errors. It shall incorporate the following fail-safe design criteria in control output logic:

(i) No false output shall result from a single point of failure in any card.
(ii) No false output shall result during power up or power down of the CSRTU.
(iii) No false output shall result from inadvertently inserting a wire circuit into a wrong I/O within the CSRTU.

5.4.5 The CSRTU shall have an internal clock for data collection coordination and time tagging. The resolution of the clock shall be one (1) millisecond or better, and the clock shall drift by no more than 100 milliseconds per day.

5.4.6 The CSRTU shall not have on-board batteries (ex. lithium batteries), as commonly used for real-time memory or internal clock time retention in the event of a power supply failure and standby battery drainage. In this respect, the CSRTU shall be capable of starting from a de-powered condition and buffering any normally time-stamped data that may take place prior to the CSRTU being time synchronized so that this data is correctly time-tagged at the CSRTU prior to transmission to Master Station.

5.4.7 The CSRTU shall support Sequence-of-events (SOE) reporting capability. To ensure that SOE data is not lost or overwritten until Master Station acknowledges receipt of the data, the CSRTU buffer shall be capable of storing a minimum of 512 events. The CSRTU shall be able to retransmit stored SOE data if requested by Master Station.

5.4.8 The CSRTU shall not use forced cooling for normal operation of its modules or components such as microprocessors and power supply units.

5.4.9 The CSRTU shall be designed with minimum periodic maintenance, preferably maintenance-free. If maintenance is required, it shall be possible to perform the maintenance work in the field without dismounting the CSRTU and without requiring the associated medium or high voltage circuit to be de-energized.

5.5 Input/Output Points

5.5.1 The CSRTU shall support Input/Output point counts (minimum) as below:

Type	Minimum Point Count
Double bit Indication	5
Single bit Indication	17
Double bit Control	5
Single bit Control	2

5.5.2 The CSRTU shall utilise the necessary sensing voltage, current limiting, input isolation, and bounce filtering for all status inputs. The debounce time period function for each status input shall be available. The input circuits of the status input modules shall be optically isolated from the external signal.

5.5.3 The wetting voltage for each input contact shall be 24 Vdc. The state of each status point shall be reported to Master Station on an exception basis. That is, a status point shall not be reported to Master Station unless the point state has changed from the last scan. The CSRTU shall also report the state of selected status points upon receipt of a demand scan request from Master Station.

As a minimum, the CSRTU shall include the following types of status input points:

(i) Single Contact, Two-State Status: For single contact, two-state status points, a single contact shall represent both states of the monitored device. One position of the contact shall indicate an alarm or failure condition, while the opposite state of the contact shall indicate the normal condition.

(ii) Double Contact, Two-State Status: For double contact, two-state status input points, separate contacts shall be provided for representing each state of the monitored device. One contact shall indicate an open condition of the monitored device. The other shall indicate a closed condition. The contacts shall be treated as a complimentary pair. Conflicting contact positions (e.g., one contact indicates CLOSED and the other indicates OPEN) shall be labelled INVALID.

5.5.4 The CSRTU shall include the following types of control points to support control actions initiated from the Master Station:

(i) On/Off Device Control: The CSRTU shall perform on/off control actions using complimentary pairs of contact outputs. One contact output shall perform the "On" control action, and a second output contact shall perform the "Off" control action. The CSRTU shall be designed such that only one output in a complimentary pair can be activated at a time.

(ii) To support the above capabilities, the CSRTU shall include momentary control outputs, as required by the feeder device being controlled. Each momentary control output shall provide a contact closure (pulse) that has programmable pulse duration. The pulse duration shall be adjustable on an individual point basis from 0.1 to 60 seconds in increments of 0.1 seconds.

(iii) Control point selection shall be cancelled if the operate command is not received within a programmable time period measured at the CSRTU by a timer. All control points shall follow a Select-Before-Operate (SBO) procedure for control operation.

5.5.5 Each CSRTU shall be equipped with a dummy breaker (latching relay) as a test indication for control function. One pair digital output in each equipped CSRTU shall be used for the TRIP and CLOSE output from the Master Station. The status of the relay shall be acquired by the RTU as a digital input for transmission to the Master Station. These indicators are included in the specified CSRTU point counts. The status of the dummy breaker shall also be indicated visually using indicators to allow visual determination of the state of the dummy breaker.

5.5.6 The CSRTU shall be equipped with a control disable switch to disconnect the power to all digital output relays or otherwise inhibit all digital output points without shutting off the RTU or otherwise affecting its operation.

5.6 Interface to SCADA Master Station

5.6.1 The CSRTU shall provide at least two (2) serial RS-232 ports, with fully configurable settings (enable/disable CTS, DTS, DCD, timeout, delays, etc). These serial ports will be used to connect to TNB's SCADA Master Station via TNB-issued Modem/Data Transmission Unit (DTU).

5.6.2 The serial port shall be individually selectable for operation at 1,200 to 115,200 bits per second, and for communications in asynchronous data formats (11 bit format data):

1) Start bit
2) 8 bit data
3) Even parity (IEC 60870-5-101 protocol) or None parity (DNP3 protocol)
4) Stop bit

5.6.3 The serial port shall be capable of supporting multiple protocols on an individually-assigned basis. As a minimum, the serial protocols shall include DNP 3.0 serial or IEC 60870-5-101.

5.6.4 All communication ports shall be internally protected against transient over-voltage by restricting "let-through" voltage below damage levels of interface circuitry.

5.6.5 The CSRTU shall also be equipped with one (1) data port for Maintenance and Configuration purposes. It shall utilise a simple configuration software (e.g Web browser or Windows 32-bit base program), to be provided and installed in TNB's personal computers by the successful Tenderer. Tools for configuration and maintenance of CSRTU (including diagnostic cables and other necessary accessories) shall also be provided.

5.7 Cabling & Installation

5.7.1 The successful Tenderer shall provide and install all interconnecting wires, cables, connectors, terminations, and other wirings required by the CSRTU.

5.7.2 Heavy duty terminal blocks with screw type terminals shall be provided for all metallic cable terminations.

5.7.3 Self-extinguishing, fire-proof vinyl marking strips shall be used to identify all external connection blocks. Marking tags shall be read horizontally. All terminals to which battery or other high voltages are connected shall be provided with fire proof covers.

5.7.4 The successful Tenderer shall ensure that the CSRTU is designed and engineered to meet the TNB's requirements fully and completely in coordination with the existing TNB's Master Station.

5.7.5 The successful Tenderer shall supply necessary materials and perform necessary fabrication, testing, wiring, and interconnection work during the process of assembling or producing the CSRTU.

5.7.6 The successful Tenderer shall complete all relevant Type Tests and Factory Acceptance Tests (FAT or Product Inspection prior to delivery.

5.7.7 The successful Tenderer shall submit relevant Engineering drawings (certified by Electrical Professional Engineer) for approval prior to fabrication. Functional test documents, operation and maintenance manuals, test procedures, test reports and any other related documents shall be submitted as and when required by TNB.